MATH 74: ALGEBRAIC TOPOLOGY

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1. Topological Spaces & Homeomorphisms

Definition 1.1. *A **topology** on a set X is a collection J of subsets of X having the following properties.*

(1) $\emptyset, X \in J$
(2) *The union of the elements of any sub-collection of J is in J.*
(3) *The intersection of finitely many elements in J is in J.*

A set for which a topology has been specified is said to be a topological space. We think of it as a pair (X, J).

Definition 1.2. *Elements of J are called **open sets** of the topological space (X, J).*

Definition 1.3. *A subspace Y of a topological space (X, J) is a subset $Y \subset X$ which is given the **subspace topology** where the open sets of $Y = \{U \cap Y : U \in J\}$*

Definition 1.4. *A space X is **path connected** if every pair of points of X can be joined by a path in X.*

Definition 1.5. *Let (X_1, J_1) and (X_2, J_2) be topological spaces. A map $f : (X_1, J_1) \to (X_2, J_2)$ is said to **continuous** if $f^{-1}(U) \in J_1$ for all $U \in J_2$.*

Definition 1.6. *Let (X, J) and (X', J') be topological spaces. Then they are **homeomorphic** if $\exists$ a continuous bijection $f : (X, J) \to (X', J')$. So that f^{-1} is also continuous.*

Exercise 1.1. Show that any two open intervals of $\mathbb{R}$ are homeomorphic (where we put subspace topology on the open intervals).

Solution. Let (a, b) and (c, d) be open intervals. **TODO:**

Remark 1.7. *Number of path connected topologies is preserved by continuous maps. For instance, $[0, 1]$ is not homeomorphic to $(0, 1)$, and $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^2$.*

Exercise 1.2.

(a) Is $\mathbb{R}^2$ homeomorphic to $\mathbb{R}^3$?
(b) Is $S^2 (= x^2 + y^2 + z^2 = 1)$ homeomorphic to $T^2 = S^1 \times S^1$?
(c) Is S^2 homeomorphic to $S^1 \times [a, b)$?

Solution.

(a) **TODO:**

Definition 1.8. *If f and f' are continuous maps from spaces X to Y we say that f is homotopic to f' if $\exists$ a continuous map $F : X \times I = [0,1] \to Y$ such that $F(x,0) = f(x)$ and $F(x,1) = f'(x)$. F is called the **homotopy** between f and f', and we write $f \simeq f'$.*

Definition 1.9. *If X and Y are topological spaces, we can define a **product topology** on $X \times Y$ by saying that a basis consists of the subsets $U \times V$ as U ranges over open sets in X and V ranges over open sets in Y.*

Definition 1.10. *Define a **path** f as $f : [0,1] \to X$ with $f(0) = x_0$ and $f(1) = x$.*

Definition 1.11. *Two paths $f, f' : [0,1] \to X$ are said to be path homotopic if there exist a continuous map $F : I \times I \to X$ such that $\forall s \in I$,*

$$F(s,0) = f(s), \qquad F(s,1) = f'(s)$$

and $\forall t \in I$,

$$F(0,t) = x_0, \qquad F(0,t) = x$$

Then we write $f \underset{p}{\simeq} f'$.

Example 1.1. *Let $f : X \to \mathbb{R}^2$ and $g : X \to \mathbb{R}^2$ then $f \simeq g$.*

Proof. Want to find F such that $F : X \times I \to \mathbb{R}^2$. $F(x,0) = f(x), F(x,1) = f'(x)$ for all $x \in X$. The following map (straight line homotopy) works.

$$F(x,t) = (1-t)f(x) + tf'(x)$$

□

Lemma 1.12. *The relations $\simeq$ and $\simeq_p$ are equivalence relations.*

Proof.

Homotopy is an equivalence relation:

(1) **Reflexive:** $f \simeq f$ by $F(s,t) = f(s)$.
(2) **Symmetric:** Let $f \simeq g$ by the homotopy $F(s,t)$. Then $g \simeq f$ is homotopic by the homotopy $G(s,t) = F(s, 1-t)$.
(3) Transitive: Let $f \simeq g$ by $F(s_1, t)$ and $g \simeq h$ by $G(s_2, t)$. We can show that $f \simeq h$ by $H(s,t) = \begin{cases} s = s_1,\ 0 \le t \le 1/2 \\ s = s_2,\ 1/2 \le t \le 1 \end{cases}$ **TODO:** check and also prove for $\simeq_p$

Thus path-homotopies are equivalence relations and we write $[f]$ to denote the path-homotopy equivalence class of f. □

Lemma 1.13 (Pasting Lemma). *Let $X = A \cup B$ be a topological space with $A, B \subset X$ are closed. Let $f : A \to Y, g : B \to Y$ be continuous and $f(x) = g(x)$, $\forall x \in A \cap B$, then*

$$h(x) = \begin{cases} f(x),\ x \in A \\ g(x),\ x \in B \end{cases}$$

is continuous.

Definition 1.14. *Let $f : I \longrightarrow X$ be a path from x_0 to x_1 and $g : I \longrightarrow X$ be a path from x_1 to x_2, then we define a new path*

$$f \star g : I \longrightarrow X$$

from x_0 to x_2 by

$$f \star g(t) = \begin{cases} f(2t),\ t \in [0, 1/2] \\ g(2t-1),\ t \in [1/2, 1] \end{cases}$$

where $\star$ is ***composition of paths****.*

Some Properties of '$\star$'.

(1) *'$\star$' is well-defined operation in the path homotopy equivalence class,*

$$[f] \star [g] = [f \star g]$$

(2) *Identity:*

$$[f] \star [e_{x_1}] = [f], \quad [e_{x_0}] \star [f] = [f]$$

where e_{x_t} is the constant path based at $x \in X$ and $t \in [0, 1]$

(3) *Inverse:*

$$[f] \star [\overline{f}] = [e_{x_0}], \quad [\overline{f}] \star [f] = [e_{x_1}]$$

where $\overline{f}(t) = f(1-t)$.

(4) *Associative:*

$$[f] \star ([g] \star [h]) = ([f] \star [g]) \star [h]$$

Proof.

(1) Let $f \simeq_p f'$ by F and $g \simeq g'$ by G, then we can show that $f \star g \simeq f' \star g'$ by defining $H : I \times I \longrightarrow X$ such that

$$H(s,t) = \begin{cases} F(2s, t),\ s \in [0, 1/2] \\ G(2s-1, t),\ s \in [1/2, 1] \end{cases}$$

(2) $f \star e_{x_1} \simeq f$ by $H : I \times I \longrightarrow X$ such that

$$H(s,t) = \begin{cases} f\left(\frac{2 \cdot s}{2-t}\right),\ s \in \left[0, \frac{2-t}{2}\right] \\ x_1,\ s \in \left[\frac{2-t}{2}, 1\right] \end{cases}$$

(3) $f \star \overline{f} \simeq_p e_{x_0}$.

Define a path $\alpha(t)$ that starts at x_0 and ends at $f(t)$. Then the path-homotopy is given by $\alpha(t) \star \overline{\alpha}(t)$.

$$H(s,t) = \begin{cases} f(2st),\ s \in [0, 1/2] \\ f(2t(1-s)),\ s \in [1/2, 1] \end{cases}$$

TODO: straight line homotopy

(4) **TODO:**

□

Remark 1.15. *Note that $\star$ operation on the set of path-homotopy equivalence classes of paths in the space X forms a groupoid structure.*

2. The Fundamental Group

Definition 2.1. *Let X be a space and $x_0 \in X$ be a point in X we say that a path $f : I \to X$ is a **loop** based at x_0 if $f(0) = f(1) = x_0$*

Definition 2.2. *The set of all path-homotopy equivalence classes of loops based at x_0 forms a group under $\star$ called the **fundamental group** of x based at x_0, denoted as $\pi_1(X, x_0)$*

- *Identity:* $[e_{x_0}]$
- *Inverses:* $[f] \to [\overline{f}]$

Example 2.1. $\pi_1(\mathbb{R}^n, (0,0)) \equiv 0 \equiv \langle e_{origin} \rangle$

Proof. Take any two loops $f(t), g(t)$ based at the origin. Then consider the straight-line homotopy

$$H(s,t) = f(s)(1-t) + tg(s)$$

TODO: (Check that it is a path homotopy) □

Definition 2.3. *Let α be a path from x_0 to x_1, where $x_0, x_1 \in X$. We define a map $\widehat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ such that*

$$\widehat{\alpha}([f]) := [\overline{\alpha}] \star [f] \star [\alpha] = [\overline{\alpha} \star f \star \alpha]$$

Proposition 2.4. *$\widehat{\alpha}$ is well-defined and an isomorphism.*

Proof. Let $f \simeq_p f'$. We need to check that

$$\begin{aligned}
[\overline{\alpha}] \star [f] \star [\alpha] &= [\overline{\alpha}] \star [f'] \star [\alpha] \\
\iff [\overline{\alpha} \star f \star \alpha] &= [\overline{\alpha} \star f \star \alpha] \\
\iff \overline{\alpha} \star f \star \alpha &\simeq_p \overline{\alpha} \star f' \star \alpha
\end{aligned}$$

Since $\star$ is well-defined, $[f] \star [\alpha] = [f'] \star [\alpha]$, $\widehat{\alpha}$ is well-defined

Let $[f], [g] \in \pi_1(X, x_0)$.

$$\begin{aligned}
\widehat{\alpha}([f]) \star \widehat{\alpha}([g]) &= [\overline{\alpha}] \star [f] \star [\alpha] \star [\overline{\alpha}] \star [g] \star [\alpha] \\
&= [\overline{\alpha}] \star [f] \star [e_{x_0}] \star [g] \star [\alpha] \\
&= [\overline{\alpha}] \star [f] \star [g] \star [\alpha] \\
&= \widehat{\alpha}([f] \star [g])
\end{aligned}$$

Hence $\widehat{\alpha}$ is a homomorphism. To check that $\widehat{\alpha}$ is an isomorphism, we see that $\widehat{\overline{\alpha}}(\widehat{\alpha})([f]) = [f]$ for all $[f] \in \pi_1(X, x_0)$. □

Corollary 2.4.1. *The fundamental group of X based at x_0 only depends on the path-connected component of $x_0 \in X$ upto isomorphism.*

Definition 2.5. *Let $h : X \to Y$ be a continuous map such that $h(x_0) = y_0$, $x_0 \in X, y_0 \in Y$. Then define $h_\star : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ by $h_\star([f]) = [h \circ f]$. This is a **push-forward** map.*

Proposition 2.6. *$h_\star$ is a well-defined homomorphism.*

Proof. Let $f \simeq_p f'$. We need to check that $h \circ f \simeq_p h \circ f'$. Let $H : I \times I \to X$ be the path homotopy between f and f'. Consider $h \circ H : I \times I \to Y$.

$$\begin{aligned} h \circ H(s,0) &= h \circ f \\ h \circ H(s,1) &= h \circ f' \\ h \circ H(0,t) &= h \circ f(0) \\ h \circ H(1,t) &= h \circ f(1) = h \circ f(0) \end{aligned}$$

Hence $h \circ H$ is the required path-homotopy.

Consider $[f], [g] \in \pi_1(X, x_0)$. We want to show that $h_\star([f]) \star h_\star([g]) = h_\star([f] \star [g])$.

$$h_\star([f]) \star h_\star([g]) = [h \circ f] \star [h \circ g]$$

Note that it suffices to show that $h(f \star g) = (h \circ f) \star (h \circ g)$. Now,

$$\begin{aligned} h(f \star g)(s) &= \begin{cases} (h \circ f)(2s), & s \in \left[0, \frac{1}{2}\right] \\ (h \circ g)(2s-1), & s \in \left[\frac{1}{2}, 1\right] \end{cases} \\ &= (h \circ f) \star (h \star g) \end{aligned}$$

Hence proved. $\square$

Theorem 2.7. *If $h : (X, x_0) \to (Y, y_0)$ and $k : (Y, y_0) \to (Z, z_0)$ then*

(a) $(k \circ h)_\star = k_\star \circ h_\star$

(b) If $\mathrm{id} : (X, x_0) \to (X, x_0)$ *is the identity map then* $(\mathrm{id})_\star$ *is the identity* $\pi_1(X, x_0) \to \pi_(X, x_0)$.

Proof. (a)

$$\begin{aligned} (k \circ h)_\star([f]) &= [(k \circ h) \circ (f)] \\ &= [k(h(f))] \\ &= [k_\star(h_\star[f])] \\ &= (k_\star \circ h_\star)[f] \end{aligned}$$

(b) $(\mathrm{id})_\star[f] = [\mathrm{id} \circ f] = [f]$

$\square$

Corollary 2.7.1. *If $h : X \to Y$ is a homeomorphism then $\pi(X, x_0) \cong \pi(Y, h(x_0)$ via $h_\star$.*

Definition 2.8. *Let $P : E \to B$ be a surjective map. A open set $U \subseteq B$ is said to be* ***evenly covered*** *by P if $P^{-1}(U) = \bigsqcup_\alpha V_\alpha$, where V_α are open sets of E for all α and $P|_{V_\alpha}$ is a homeomorphism of V_α onto U.*

Definition 2.9. *Let $P : E \to B$ be surjective if every point $b \in B$ has a neighbourhood $U \subset B$ such that U is evenly covered by P. Then we say that P is a* ***covering map*** *and E is a* ***covering space*** *of B.*

Example 2.2.

(1) id $: X \to X$.
(2) $X \times \{1, 2, \cdots, n\} \xrightarrow{P} X$ *where* $P(X, i) = x$.

Theorem 2.10. *The map* $P : \mathbb{R} \to S'$ *such that* $x \mapsto (\cos(2\pi x), \sin(2\pi x))$ *is a covereing space.*

Proof. $P^{-1}(U) = \bigsqcup_{n \in \mathbb{Z}} \left(n - \frac{1}{4}\right) \left(n + \frac{1}{4}\right) =: V_n$

Restricted to each V_n the map P is a homeomorphism because $\sin(2\pi x)$ is strictly monotonic on V_n for all n.

TODO: finish $\square$

Definition 2.11. *If a path-connected space* X *has* $\pi_1(X, x_0) \cong 0$ *then we say that* X *is* ***simply-connected***.

Lemma 2.12. *If* X *is simply-connected, then any two paths in* X *with the same initial point and end point are path-homotopic.*

Proof. $f \star \overline{g} \simeq_p e_{x_0}$ and $f \simeq_p f \star e_{x_1} \simeq_p f \star (\overline{g} \star g) \simeq_p (f \star \overline{g}) \star g \simeq_p e_{x_0} \star g \simeq_p g$. $\square$

Definition 2.13. *Let* $p : E \to B$ *be a surjective map,* $f : X \to B$ *be a map from* $X \to B$*. Then a map* $\widetilde{f} : X \to E$ *is called* ***lifting*** *of* f *to* E *if* $p \circ \widetilde{f} = f$.

$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, dashed, "\widetilde{f}"] \arrow[r, "f"'] & B \end{tikzcd}$$

Lemma 2.14 (Path Lifting Lemma)**.** *Let* $p : E \to B$ *be a covering map with* $p(e_0) = b_0$*. Any path* $f : [0, 1] \to B$ *beginning at* b_0 *can be lifted uniquely to a path* $\widetilde{f} : [0, 1] \to E$ *beginning at* e_0*.*

Proof. Cover B by open sets U, each of which is evenly covered by p. We can find a subdivision (partition) defined by $s_0, \cdots, s_n \in [0, 1]$ such that $f([s_i, s_{i+1}])$ is contained entirely inside a evenly covered U for all i. **TODO:** Think about why this is possible. Now we will define $\widetilde{f}$ inductively. Suppose that $\widetilde{f}(0) = e_0$ and it has been defined for $0 \le s \le s_i$ for some i. We will now define $\widetilde{f}$ at $[s_i, s_{i+1}]$. Suppose that $f([s_i, s_{i+1}]) \subset U$ where U is evenly covered open set of B. This implies that $p^{-1}(U) = \sqcup_\alpha V_\alpha$. Now, $\widetilde{f}(s_i)$ lies in one such V_α say V_0.

Define $\widetilde{f}$ on $[s_i, s_{i+1}]$ as $\widetilde{f}(s) := (p|_{V_0})^{-1}(f(s))$ for all $s \in [s_i, s_{i+1}]$. By pasting lemma $\widetilde{f}$ is continuous on $[0, s_{i+1}]$ and we argue inductively to define $\widetilde{f}$ for $[0.1]$.

By induction, we can again suppose that for another lift $\widetilde{f}' : [0, 1] \to E$ of f with $\widetilde{f}'(0) = e_0$, it agress with $\widetilde{f}$ of $0 \le S \le S_i$ for some i

$$\widetilde{f}'(s) = \widetilde{f}(s), \qquad \forall s \in [0, s_i]$$

Now we would like to show that

$$\widetilde{f}'(s) = \widetilde{f}(s), \qquad \forall s \in [s_i, s_{i+1}]$$

$$\widetilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s), \qquad \forall s \in [s_i, s_{i+1}]$$

$\widetilde{f}'(s_i)$ lies in V_0 since $\widetilde{f}'(s_i) = \widetilde{f}(s_i)$. $\implies$ $\widetilde{f}'([s_i, s_{i+1}])$ lies in V_0.

$p \circ \widetilde{f}' = f$ on $[s_1, s_{i+1}]$. But $\widetilde{f}'([s_i, s_{i+1}]) \subset V_0$ and $P|_{V_0}$ is a homeomorphism.

Then for all $s \in [s_i, s_{i+1}]$

$$\implies \widetilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s) = \widetilde{f}(s)$$

□

Lemma 2.15 (Path Homotopy Lifting Lemma)**.** *Let $p : E \to B$ be a covering map. Let $p(e_0) = b_0$. Let $f : I \times I \to B$ be a map, with $f(0,0) = b_0$. Then there exists a lifting of f to E such that $\widetilde{f}(0,0) = e_0$. If f is a path homotopy, so is $\widetilde{f}$.*

Proof. **TODO:** $I \times I$ is a rectangle. Pick a parition of both sides, then the image of these small rectangles will be contained in U

1. Define $\widetilde{f}$ on $0 \times I$ and $I \times 0$ by lifting both of those paths to paths that begin at e_0.

2. Take a subdivision of $I \times I$ say $I_i \times J_j$ such that each of those small rectangles are contained in an evenly covered open set of B.

3. By induction assume that $\widetilde{f}$ has been defined on $I_i \times J_j$ such that $i < i_0, j = j_0$.

4. Now, to define $\widetilde{f}$ on $I_{i_0} \times I_{j_0}$ we consider the image of the union of left and bottom edge of $I_{i_0} \times I_{j_0}$ under $\widetilde{f}$. This lies inside $V_0 \subset p^{-1}(U)$. □

Remark 2.16. *The lifted path homomoty is also unique.*

Lemma 2.17. *Let $P : E \to B$ be a covering map with $P(e_0) = b_0$. Let f, g be two paths in B from b_0 to b_1. Let $\widetilde{f}, \widetilde{g}$ be their lifts beginning at e_0. Then if $f \simeq_p g \implies \widetilde{f}$ and $\widetilde{g}$ end at the same point and they are path homotopic.*

Proof. Assume $f \simeq_p g$ with the path homotopy $F : I \times I \to B$. Then Lemma 2.15 implies that F lifts to $\widetilde{F} : I \times I \to E$ with $\widetilde{F}(0,t) = e_0$.

Then we can say that $\widetilde{F}(1,t) = e_1$ for some $e_1 \in P^{-1}(b_1)$, $\forall t \in I$. By uniqueness of path lifting, f, g have unique lifts to E, beginning at e_0. We also know that $\widetilde{F}(s,0)$ and $\widetilde{F}(s,1)$ are lifts of f, g both beginning at e_0.

Then $\widetilde{F}(s,0) = \widetilde{f}$ and $\widetilde{F}(s,1) = \widetilde{g}$ for all $s \in I$. Hence $\widetilde{F}(1,0) = \widetilde{F}(1,1) \implies \widetilde{f}(1) = \widetilde{g}(1)$, and $\widetilde{f} \simeq \widetilde{g}$. □

Theorem 2.18. $\pi_1(S^1, (1,0) = b_0) \cong (\mathbb{Z}, +)$

Proof.

$$\begin{tikzcd}
 & \mathbb{R} \arrow[d] \\
I \arrow[ur, "\widetilde{f}"] \arrow[r, "f"] & S^1
\end{tikzcd}$$

Consider the convering map

$$P : \mathbb{R} \to S^1,\ P(x) = (\cos(2\pi x), \sin(2\pi x))$$

Let $[f] \in \pi_1(S^1, b_0)$. Let $\widetilde{f}$ be the lift of f to $\mathbb{R}$ beginning at 0. Now consider $\widetilde{f}(1)$. It will be an integer since $\widetilde{f}(1) \in P^{-1}(b_0) = P^{-1}((1,0))$. Define

$$\phi : \pi_1(S^1, b_0) \to \mathbb{Z}, \quad \phi([f]) = f(1)$$

By previous lemma, ϕ is well-defined.

Now we show that ϕ is surjective. Let $n \in \mathbb{Z}$. Define $\widetilde{f}$ to be the path in $\mathbb{R}$ which starts at 0 and ends at n. Then by definition, $(P \circ \widetilde{f})$ will start at $(1,0)$ and end at $(1,0)$ where $\widetilde{f}$ is by definition the lift of $(P \circ \widetilde{f})$ beginning at 0. So $\phi([P \circ \widetilde{f}]) = n$.

To show that ϕ is injective suppose that $\phi([f]) = \phi([g]) = n$ for some $[f], [g] \in \pi_1(S^1, b_0)$ we need to show that $[f] = [g] \in \pi_1(S^1, b_0)$.

Now let $\widetilde{f}$ and $\widetilde{g}$ be two lifts of f and g respectively beginning at 0. Moreover, since $\mathbb{R}$ is simply connected and $\widetilde{f}(0) = \widetilde{g}(0) = 0$, $\widetilde{f}(1) = \widetilde{g}(1) = n$, $\widetilde{f} \simeq_p \widetilde{g}$, then $P \circ \widetilde{F}$ will be the path-homotopy between f and g. This implies $[f] = [g]$.

Now we want to show that ϕ is a group homomorphism. Let $[f], [g] \in \pi_1(S^1, b_0)$ such that $\phi([f]) = n$, $\phi([g]) = m$ for $n, m \in \mathbb{Z}$. Now we want to show that $\phi([f] \star [g]) = \phi([f \star g]) = n + m$. We need to find a lift of $f \star g$ beginning at 0 and ending at $n + m$.

Let $\widetilde{f}, \widetilde{g}$ be lifts of f, g respectively, $\widetilde{f}(0) = \widetilde{g}(0) = (0)$.

Define a path on $\mathbb{R}$ by

$$h(s) = \begin{cases} \widetilde{f}(2s), & s \in \left[0, \frac{1}{2}\right] \\ n + \widetilde{g}(2s-1), & s \in \left[\frac{1}{2}, 1\right] \end{cases}$$

Then h is a path beginning at 0 and ending at $n + m$. Now consider,

$$P(h(s)) = \begin{cases} P \circ \widetilde{f}(2s), & s \in \left[0, \frac{1}{2}\right] \\ P \circ (n + \widetilde{g}(2s-1)) = P \circ \widetilde{g}(2s-1), & s \in \left[\frac{1}{2}, 1\right] \end{cases}$$

This implies that $h(s)$ is a lift of $f \star g(s) \implies \phi([f \star g]) = h(1) = n + m$ $\square$

Corollary 2.18.1. *Fundamental Theorem of Algebra.*

Proof. Let $p(z) = z^n + a_1 z^{n-1} + \cdots + a_n$, where $a_i \in \mathbb{C}$.

For the sake of contradiction assume that p has no roots in $\mathbb{C}$. Then for each $r \in \mathbb{R}^+$ define a function

$$f_r(s) = \frac{\frac{p(re^{2\pi i s})}{p(r)}}{\left|\frac{p(re^{2\pi i s})}{p(r)}\right|} \in \mathbb{C}$$

Note that $f_r(s)$ is a loop in S^1 based at $(1,0)$ for all r. ($f_r(0) = (1,0) = f_r(1)$ and $|f_r(s)| = 1, \forall s$). Note that as r varies f_r gives a path homotopy in S^1 between loops that are based at $(1,0)$. Now, note that $f_0(s)$ is the trivial loop based at $(1,0)$. This implies,

$$[f_r(s)] = [f_0(s)] = 0 \in \pi_1(S^1,(1,0))$$

Now fix a large value of r such that $r > |a_1| + \cdots |a_n|$ and $r > 1$. Now, we have for $|z| = r$

$$|a_1 z^{n-1} + \cdots + a_n| \leq |a_1| r^{n-1} + \cdots + |a_n|.$$

Thus if $r > |a_1| + \cdots + |a_n|$, then

$$|z^n| = r^n > r^{n-1}(|a_1| + \cdots + |a_n|) \geq |a_1 z^{n-1} + \cdots + a_n|.$$

Hence,

$$|z^n| > |a_1 z^{n-1} + \cdots + a_n|$$

This implies that the polynomial $p_t(z) = z^n + t(a_1 z^{n-1} + \cdots + a_n)$, $\forall t \in [0,1]$ has no roots in circle $|z| = r$. If we now replace p by p_t in the definition of $f_r(s)$ we get

$$f_r(s,t) = \frac{\frac{p_t(re^{2\pi is})}{p_t(r)}}{\left|\frac{p_t(re^{2\pi is})}{p_t(r)}\right|}$$

At $t = 1$,

$$f_r(s,1) = f_r(s) \in \pi_1(S^1,(1,0))$$

At $t = 0$,

$$f_r(s,0) = e^{2\pi ins}, \quad \forall s \in [0,1]$$

So $[f_r(s,0)] = n \in \mathbb{Z} \cong \pi_1(S^1,(1,0))$. But $f_r(s,t)$ as we vary $t \in [0,1]$ gives a path homotopy between $f_r(s,0)$ and $f_r(s,1) \implies [f_r(s,0)] = [f_r(s)] \in \pi_1(S^1,(1,0)) \implies n = 0$ contradicting that p is not constant. □

Remark 2.19. *In general, we have $P : (E, e_0) \to (B, b_0)$ be a covering map. If E is path-connected then there exists a surjection $\phi : \pi_1(B, b_0) \to P^{-1}(b_0)$. If E is simply-connected, then ϕ is a bijection.*

Proof. HW1 iirc □

Theorem 2.20.

$$\pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0) \cong \mathbb{Z} \cong \pi_1(S^1), \quad b_0 \in S^1.$$

Proof. Let $x_0 \in S^1$. Then the inclusion map

$$J : (S^1, b_0) \to (\mathbb{R}^2 \setminus \{(0,0)\}, b_0)$$

induces an isomorphism

$$J_* : \pi_1(S^1, b_0) \to \pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0).$$

Consider

$$r : \mathbb{R}^2 \to S^1, \quad r(x) = \frac{x}{\|x\|}.$$

Observe that $r \circ J = \text{id}$, thus

$$r_* \circ J_* = \text{id}.$$

Thus J_* is injective.

Choose $[f] \in \pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0)$. Recall that,

$$J_* \circ r_*([f])$$

is explicitly $J \circ r \circ f$. Let

$$g(s) = \frac{f(s)}{\|f(s)\|}.$$

Next show $g \simeq_p f$. Define

$$F : I^2 \to \mathbb{R}^2 \setminus \{(0,0)\}$$

such that

$$F(s,t) = t \cdot \frac{f(s)}{\|f(s)\|} + (1-t)f(s).$$

F is the straight line homotopy and does not cross 0.

Since $f(s) \neq 0$ by definition, suppose

$$t \cdot \frac{1}{\|f(s)\|} + (1-t) = 0$$

implies $\|f(s)\|$ negative or $t \notin [0,1]$.

Thus F is well-defined, since $f \simeq_p g$,

$$J_* \circ r_*([f]) = [f].$$

Thus surjection. □

3. Retractions

Definition 3.1. *If A is a subspace of X s.t. $\exists$ a map*

$$r : X \to A$$

with $r(a) = a$ for all $a \in A$, then we say A is a retract of X and r is a retraction map from X to A.

Definition 3.2. *Let A be a subspace of X. A is a deformation retract of X if there exists a homotopy*

$$H : X \times I \to X$$

such that

$$\begin{gathered} H(x,0) = x \quad \forall x \in X, \\ H(x,1) \in A, \\ H(a,t) = a. \end{gathered}$$

Theorem 3.3. *Let A be a deformation retract of X. Then $j : (A, x_0) \hookrightarrow (X, x_0)$ induces an isomorphism in $\pi_1(A, x_0)$ to $\pi_1(X, x_0)$. If A in a retract of X, then j is injective.*

Proof. **TODO:** hw ☐

Remark 3.4. *If X deformation retracts to A. then $\pi_1(X) \cong \pi_1(A)$.*

Remark 3.5. *Let $\iota : A \hookrightarrow X$ then $\iota_\star : \pi_1(A) \to \pi_1(X)$ is injective if X only retracts to A.* ***TODO:*** *only retracts??*

4. APPLICATIONS OF π_1-COMPUTATIONS

Theorem 4.1 (Brouwer-Fixed Point Theorem)**.** *Every map $h : D^2 \to D^2$ has a fixed point, i.e. $h(x) = x$ for at least one $x \in D^2$.*

Proof. Suppose towards contradiction $h(x) \neq x$ for all $x \in D^2$. Define a function $\gamma : D^2 \to S^1$ as follows: consider the straight line starting at $h(x)$ and going to x. Define $r(x)$ as the intersection of that line with $\delta D^2 = S^1$. Now r is a continuous function, and $r|_{S^1} = Id_{S^1}$. That is, r is a retraction of D^2 to S^1.

Since for the inclusion $\iota : S^1 \hookrightarrow D^2$ we should have $\pi_1(S^1, (1, 0)) \xrightarrow{\iota_\star} \pi_1(D^2, (1, 0))$ is an injection which gives a contradiction. □

Theorem 4.2 (Borsuk-Ulam Theorem)**.** *Every map $f : S^2 \to \mathbb{R}^2$ has a pair of anti-podal points (x is an antipode of $-x$, $x \in S^2$) in S^2 such that $f(x) = f(-x)$.*

Proof. Suppose that this is false. Then $f(x) \neq f(-x)$ for all $x \in S^2$. We define $g : S^2 \to S^1$ by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

Define a loop η by the equator that lies in the $X - Y$ plane of S^2.

$$\eta(S) = (\cos(2\pi s), \sin(2\pi s), 0),\ s \in [0, 1]$$

and let $h : I \to S^1$ defined by $h := g \circ \eta$. Note that since $g(x) = -g(-x)$ we have $h(s + \frac{1}{2}) = -h(s)$ for all $s \in \left[0, \frac{1}{2}\right]$.

$$\begin{aligned} g(\cos(2\pi(s + 1/2)), \sin(2\pi(s + 1/2)), 0) &= g(-\cos(2\pi s), -\sin(2\pi s), 0)) \\ &= -g(\cos(2\pi s), \sin(2\pi s), 0) \end{aligned}$$

By path-lifting lemma, we can lift h to a path in $\mathbb{R}$, say $\widetilde{h}$ starting at 0 such that

$$\widetilde{h}\left(s + \frac{1}{2}\right) = \widetilde{h}(s) + \frac{q(s)}{2}$$

for some odd integer values $q(s)$. But now, $q(s) = 2(\widetilde{h}\left(s + \frac{1}{2}\right) - \widetilde{h}(s))$ and hence it is also a continuous function. This implies $q(s)$ is a constant map (since the domain of q is conected), i.e. $\widetilde{h}\left(s + \frac{1}{2}\right) = \widetilde{h}(s) + \frac{q}{2}$.

$$\widetilde{h}(1) = \widetilde{h}\left(\frac{1}{2}\right) + \frac{q}{2} = \widetilde{h}(0) + q$$

This implies $\widetilde{h}(1) = q$ and hence $[h] \in \pi_1(S^1, (1, 0))$ is not homotopic to the identity. But since η is nullhomotopic in S^2, $g \circ \eta$ is homotopic to a constant map $[h] = 0 \in \pi_1(S^1, (1, 0))$. A contradiction! □

5. Homotopy Equivalences

Theorem 5.1. *$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$ for any pair of topological spaces X and Y.*

Proof. Let $P_1 : X \times Y \to X$ and $P_2 : X \times Y \to Y$ be the respective projections.

$$\phi : \pi_1(X \times Y, x_0 \times y_0) \to \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

where

$$\phi([f]) = (P_1)_\star[f] \times (P_2)_\star[f]$$

ϕ is surjective: Let $[g] \in \pi_1(X, x_0)$ and $[h] \in \pi_1(Y, y_0)$. Then define $[g] \times [h] : I \to X \times Y$ by $s \mapsto g(s) \times h(s)$. Then $\phi([g \times h]) = [g] \times [h]$.

ϕ is injective: Suppose that $[f] \in \pi_1(X \times Y, x_0 \times y_0)$ such that $\phi([f]) = (P_1)_\star([f]) \times (P_2)_\star([f]) = 0$. Then $P_1 \circ f \simeq_p e_{x_0}$ and $P_2 \circ f \simeq_p e_{x_1}$ by homotopies F_t ans G_t.

Then, $F_t \times G_t : I \to X \times Y$ gives $f \simeq_p e_{x_0 \times y_0} \implies [f] = 0$.

$\implies \pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}$. □

Definition 5.2. *We say $f : X \to Y$ is a homotopy equivalence if there exists a $g : Y \to X$ such that $f \circ g \simeq id_Y$ and $g \circ f \simeq id_X$. Then we say that X is homotopy equivalent to Y and write it as $X \simeq_h Y$.*

Remark 5.3. *Homotopy equivalence is an equivalence relation on topological spaces.*

Proposition 5.4. *Suppose that X deformation retracts to A. Then $X \simeq_h A$.*

Proof. Let $H_t : X \to A$ be the deformation retraction. Then consider $H_1(X) \subseteq A$, and $H_1|_A = id$. Let us call $r := H_1$ and let $\iota : A \hookrightarrow X$ be the inclusion map. Then $r \simeq id \implies \iota \circ r \simeq id$ and $r \circ \iota = id \implies X \simeq_h A$. □

Remark 5.5. *Note not all homotopy equivalence come from deformation retractions.*

Remark 5.6. *π_1 is not only an invariant of homeomorphism class of a topological space but also an invariant of the homotopy equivalence class of the space.*

Theorem 5.7. *Let $\phi : X \to Y$ be a homotopy equivalence. Then $\phi_\star : \pi_1(X, x_0) \to \pi_1(Y, \phi(x_0))$ is an isomorphism.*

Lemma 5.8. *If $\phi_t : X \to Y$ be a homotopy and h is the path given by $\phi_t(x_0)$ for a choice of $x_0 \in X$, then we have $(\phi_0)_\star = \widehat{\overline{h}}((\phi_1)_\star)$ on $\pi_1(X, x_0)$. That is,*

$$\begin{tikzcd}
\pi_1(X, x_0) \arrow[r, "(\phi_1)_\star"] \arrow[dr, "(\phi_0)_\star"'] & \pi_1(Y, \phi_1(x_0)) \arrow[d, "\widehat{\overline{h}}"] \\
& \pi_1(Y, \phi_0(x_0))
\end{tikzcd}$$

Proof. Let $[f] \in \pi_1(X, x_0)$. Let h_t be a reparameterization of the path h such that $h_t(s) := h(st)$. In particular, h_t is a path from $\phi_0(x_0)$ to $\phi_t(x_0)$. Now the composition, $h_t \star \phi_t \circ f \star \overline{h}_t$ gives a path homotopy between

$$h_0 \star \phi_0 \circ f \star \overline{h}_0 = e_{x_0} \star \phi_0 \circ f \star \overline{e}_{x_0} \simeq \phi_0 \circ f$$

and

$$h_1 \star \phi_1 \circ f \star \overline{h}_1 = h \star \phi_0 \circ f \star \overline{h} \implies (\phi_0)_\star = \widehat{\overline{h}}((\phi_1)_\star)$$

$\square$

Theorem 5.9. *Let $\phi : X \to Y$ be a homotopy equivalence. Then $\phi_\star : \pi_1(X, x_0) \to \pi_1(Y, \phi(x_0))$ is an isomorphism.*

Proof. Since ϕ is a homotopic equivalence, there exists a $\psi : Y \to X$ such that $\phi \circ \psi \simeq \mathrm{id}_Y$ and $\psi \circ \phi \simeq \mathrm{id}_X$. Consider,

$$\pi_1(X, x_0) \xrightarrow{\phi_\star} \pi_1(Y, \phi(x_0)) \xrightarrow{\psi_\star} \pi_1(X, \psi(\phi(x_0))) \xrightarrow{\phi_\star} \pi_1(Y, \phi(\psi(\phi(x_0))))$$

Note that $\psi_\star \circ \phi_\star = (\psi \circ \phi)_\star = \widehat{\overline{h}} \circ (\mathrm{id}_X)_\star$ for some path h in X (by the previous lemma). Tha implies $\psi_\star \circ \phi_\star = \widehat{\overline{h}} \implies \psi_\star$ is surjective since $\widehat{\overline{h}}$ is an isomorphism.

$\psi_\star$ is injective because

$$\phi_\star \circ \psi_\star = \widehat{\overline{h'}} \circ (\mathrm{id}_Y)_\star$$

for some path h' in Y (by the previous lemma).

Hence, $\phi_\star = (\psi_\star)^{-1} \circ \widehat{\overline{h}}$ which is an isomorphism. $\square$

6. $\pi_1(S_n)$

Lemma 6.1. *If a space X is the union of a collection of path-connected open sets A_α each containing the base point $x_0 \in X$. Let $A_\alpha \cap A_\beta$ be path connected for any α, β, then every loop in X based at x_0 is path homotopic to a composition of loops each contained in a single A_α for some α and based at x_0.*

Proof. Let $[f] \in \pi_1(X, x_0)$. Suppose that $0 = S_0 < S_1 < \ldots < S_m = 1$ is a subdivision of $[0,1]$ such that $f\mid_{[S_{i-1}, S_i]}$ is contained in a single A_α for all i and some α dependening on i. Let us denote that A_α by A_i.

Note $f(0) = x_0$. Define g_i to be the path from x_0 to $f(S_i) \in A_i \cap A_{i+1}$.

Consider,

$$(f \star \overline{g_1}) \star (g_1 \star f \star \overline{g_2}) \star (g_2 \star f \star \overline{g_3}) \star \cdots \star (g_{m-1} \star f) \simeq_p f$$

while each of them are loops based at x_0 in A_is as i varies. □

Theorem 6.2. *$\pi_1(S^n, x_0) \cong \{0\}$ if $n \geq 2$.*

Proof. On S^2 consider two open sets $A_1 = S^2 \backslash \{\text{north pole}\}$ and $A_2 = S^2 \backslash \{\text{south pole}\}$. Let $[f] \in \pi_1(S^2, x_0)$. Then by the lemma $f \simeq_p g_1 \star g_2$ where $[g_1] \in \pi_1(A_1, x_0)$ and $[g_2] \in \pi_1(A_2, x_0)$ for some g_1 and g_2 loops based at x_0. Then,

$$[f] = 0 \in \pi_1(S^2, x_0) \implies \pi_1(S^2, x_0) = 0$$

□

Remark 6.3 (Stereographic Projection)**.** *$S^n \setminus \{$ some pt $\}$ is homeomorphic to $\mathbb{R}^n$.*

7. Covering Spaces (again)

Definition 7.1. ***TODO:*** $\mathbb{R}P^n := S^n / \{x \sim -x\}$ *(quotinent space)*

Theorem 7.2. *Define $p : S^n \to \mathbb{R}P^n$, $x \mapsto [x]$. Then p is a covering map.*

Proof. First show that p is an open map (i.e. U open $\implies$ $p(U)$ open). Let $V \subset S^n$ be an open set (S^n has subspace topology from $\mathbb{R}^{n+1}$). Consider 'antipodal map',

$$a : S^n \to S^n, \quad x \mapsto -x$$

Then a is a homeomorphism. Thus its inverse is continuous, and $a(V)$ is open in S^n. Now since $p^{-1}(p(V)) = V \cup a(V)$, which is open in S^n, p is an open map (using definition of quotient topology).

Now we will show p is a covering map. Let $y \in \mathbb{R}P^n$. Pick $x \in p^{-1}(y)$. Choose a small neighborhood U of x in S^n such that U contains no antipodal points. Then $p|_U : U \to p(U)$ is a homeomorphism (since p is open and continuous and a bijection).

Similarly, $p|_{a(U)} : a(U) \to p(a(U))$ is also an homeomorphism. Note $p(a(U)) = p(U)$ which is an open set around $y \in \mathbb{R}P^n$. So each point in $\mathbb{R}P^n$ is evenly covered by an open set in $\mathbb{R}P^n$. Hence, p is a covering map (specifically a 2-fold covering).

Then $\pi_1(\mathbb{R}P^n, x_0) \leftrightarrow p^{-1}(x_0) \implies |\pi_1(\mathbb{R}P^n, x_0)| = 2 \implies |\pi_1(\mathbb{R}P^n, x_0)| = \mathbb{Z}/2\mathbb{Z}$. □

Corollary 7.2.1. *$\mathbb{R}P^n$ is not homotopy equivalent to S^n for $n \geq 2$.*

Proof. Fundamental group is a homotopy-equivalence invariant, and S^n for $n \geq 2$ is simply connected. □

Remark 7.3. *Homeomorphism $X \cong Y \implies X \simeq_h Y$ (homotopy equivalence).*

Theorem 7.4. *$\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.*

Proof. Suppose for contradiction there exists a homeomorphism f from $\mathbb{R}^2 \to \mathbb{R}^n$. Then $f\mid_{\mathbb{R}^2 \setminus \{\text{pt}\}}: \mathbb{R}^2 \setminus \langle x \rangle \to \mathbb{R}^n \setminus \{f(x)\}$ is also a homeomorphism. So $\mathbb{R}^2 \setminus \{x\} \simeq_h \mathbb{R}^n \setminus \{f(x)\}$ must be homotopy equivalent, But π_1 is a homotopy equivalence so we should have $\pi_1(\mathbb{R}62 \setminus \{x\}) \cong \pi_1(\mathbb{R}^n \setminus \{f(x)\}) \implies \mathbb{Z} \cong 0$. A contradiction! □

Theorem 7.5. *Fundamental group of the 'figure eight space' is not abelian.*

Proof. Let X be the figure eight space. Then $X = S^1 \sqcup S^1 \setminus (1,0)\widetilde{(}1,0)$ with the quotient space topology. Let x_0 be the base point. Define a covering space $E \subseteq \mathbb{R}^2$, where $E = \{\text{x-axis}\} \cup \{\text{y-axis}\} \cup \{\text{one circle tangent to each the x-axis and y-axis at each nonzero integer point}\}$ (fig 2.1 in munkres). We claim that there exists a projection map $p : E \to X$. Consider two paths $\widetilde{f}, \widetilde{g} : I \to E$ defined by $\widetilde{f}(s) = (s,0)$, $\widetilde{g}(s) = (0,s)$. Define $f := p \circ \widetilde{f}$, $g := p \circ \widetilde{g}$. That is f wraps once around A, g wraps once around B.

Claim: $[f \star g] \neq [g \star f] \in \pi_1(\infty, x_0)$.

Proof: Claim $\iff$ $f \star g \not\simeq_p g \star f$. For the sake of contradiction suppose $f \star g \simeq_p g \star f$. Now consider the lift of $f \star g$ beginning at $(0,0)$ to E given by $\widetilde{f} \star B_1$, and the lift of $g \star f$ beginning at $(0,0)$ to E given by $\widetilde{g} \star A_1$. If they are path-homotopic, they should have the same end points, but $(\widetilde{f} \star B_1)(1) = (1,0) \neq (0,1) = (\widetilde{g} \star A_1)(1)$. □

Corollary 7.5.1. *$\pi^2 \# \pi^2 \ncong S^2 . \mathbb{R}P^2$, where π^2 is the hollow torus, and # is the connected sum.*

Proposition 7.6. *Let $h : S^1 \to Y$ where Y is some topological space, then h is homotopic to a constant map iff h can be extended to a continuous map $D^2 \to Y$.*

Proof. ($\implies$)

Let $H : S^1 \times I \to Y$ be a homotopy from h to a constant map $S^1 \to x_0$. Define $\pi : S^1 \times I \to D^2$ by $\pi(x,t) = (1-t)x$. Then $\pi\mid_{S^1 \times [0,1)}$ is a bijection to $D^2 \setminus \{0\}$ and it maps $S^1 \times \{1\}$ to $0 \in D^2$. Since $S \times I$ is compact and D^2 is Hausdorff, π is a closed map. Now H induces a map $g : D^2 \to Y$ such that $g \circ \pi = H$, given by

$$g(x) = \begin{cases} H(\pi^{-1}(x)) & x \neq 0 \\ x_0 & x = 0 \end{cases}$$

Then g will be the extension we want ... □

TODO: 05/04 notes

8. Classification of Covering Space

Definition 8.1. *We say that a covering space $\widetilde{X}$ of X is a universal cover if $\pi_1(\widetilde{X}) = 0$.*

Example 8.1.

1. *$\mathbb{R}$ is a universal cover of S^1.*
2. *Every simply connected space is a universal cover of itself.*
3. *$\mathbb{C} \to \mathbb{C}$ where $z \mapsto z^2$.*
4. *S^n is a universal cover of $\mathbb{R}P^n$ for $n \geq 2$*

Remark 8.2. *An object with property x is universal if any other object with property x factors through the first object.*

Theorem 8.3. *Suppose that $\widetilde{X}$ is a universal cover of X, then if X is path-connected and locally-connected, any other path-connected cover X_1 of X, $\widetilde{X}$ covers X_1. Moreover the following diagram commutes.*

$$\begin{tikzcd} \widetilde{X} \arrow[r, dashed, "p_1"] \arrow[rr, bend right, "p"'] & X_1 \arrow[r, "p_2"] & X \end{tikzcd}$$

Proof. The proof follows from the following two lemmas. □

Lemma 8.4. *Suppose that $X, X_1, \widetilde{X}$ are all path-connected such that $X_1 \xrightarrow{p_1} X$, $\widetilde{X} \xrightarrow{p} X$ are covers with $\pi_1(\widetilde{X}) = 0$ then $\exists\, p_2 : \widetilde{X} \to X_1$ such that $p_1 \circ p_2 = p$.*

Proof. Follows from the general lifting proposition. □

Lemma 8.5. *Let $X, X_1, \widetilde{X}, p, p_1, p_2$ be defined as before, then p_2 is a covering map.*

Proof. Let $x_1 \in X_1$ be a point, U be a neighborhood of $p_1(x_1) \in X$ that is evenly covered by both p_1 and p. Now let $p^{-1}(U) = \sqcup_\alpha U_\alpha$ and $p_1^{-1}(U) = \sqcup_{\alpha'} V_{\alpha'}$. Moreover assume that V is in the slice of $p_1^{-1}(U)$ where x_1 lies. Claim: V is evenly covered by p_2. Proof: Consider the U_α's in $\widetilde{X}$ that are mapped to V. Lets say that collection is $\{U_\beta\}$. Then $p_2^{-1}(V) = \sqcup_\beta U_\beta$ and $p_2|_{U_\beta}$ is a homeomorphism since it is a composition of $p|_{U_\beta}$ and $(p_1|_V)^{-1}$ which are both homeomorphisms □

Theorem 8.6. *Let $\widetilde{X}$ be a universal cover of X, then $\widetilde{X}$ is unique upto homeomorphism.*

Proof. Hint: Lifts are unique if we fix a base point of the lift.

Let $\widetilde{X}$ and $\overline{X}$ be two universal covers of X. Then,

$$\begin{tikzcd} & (\overline{X}, \overline{x_0}) \arrow[d, "p_2"] \\ (\widetilde{X}, \widetilde{x_0}) \arrow[ur, "f"] \arrow[r, "p_1"] & (X, x_0) \end{tikzcd} .$$

$$\begin{tikzcd} & (\widetilde{X}, \widetilde{x_0}) \arrow[d, "p_1"] \\ (\overline{X}, \overline{x_0}) \arrow[ur, "g"] \arrow[r, "p_2"] & (X, x_0) \end{tikzcd} .$$

$$p_1 = p_2 \circ f, \qquad p_2 = p_1 \circ g$$

That is,

$$p_1 = p_1 \circ g \circ f$$

Moreover, $g \circ f(\widetilde{x_0}) = \widetilde{x_0}$. Hence, $g \circ f$ is a lift of p_1 to $\widetilde{X}$ by p_1. Identity is also such a lift, then $g \circ f = \mathrm{id}$ (by uniqueness of lift homotopy). $\square$

TODO: Read this proof ?? For any sub space of $\pi_1(X) \leftrightarrow$ cover X_1 of X with $p_\star(\pi_1(X_1)) \cong$ that subspace.

Remark 8.7. *Any collection of homomorphisms $\phi_\alpha : G_\alpha \to H$ extends uniquely to $\phi : \star_\alpha G_\alpha \to H$ such that $\phi(g_1 \cdots g_n) \mapsto \phi_{\alpha_1}(g_1)\phi_{\alpha_2}(g_2) \ldots \phi_{\alpha_n}(g_n)$.*

where G_α is a free group, H is a group, and $\star_\alpha G_\alpha$ consists of all words of arbitrary length where each letter belongs to a G_{α_i} and consequetive letters are from different G_{α_i}.

Now suppose that X is decomposed as a union of path-connected spaces $\{A_\alpha\}$ each of which contains the base point $x_0 \in X$. Then inclusion $A_\alpha \hookrightarrow X \implies j_\alpha : \pi_1(A_\alpha) \to \pi_1(X)$ induce $\Phi : \star_\alpha \pi_1(A_\alpha) \to \pi_1(X)$.

Let $(\iota_{\alpha\beta})_\star : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(A_\alpha)$ be induced from inclusion. Hence,

$$(j_\alpha)_\star \circ (\iota_{\alpha\beta})_\star = (j_\beta)_\star \circ (\iota_{\beta\alpha})_\star : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(X)$$

If $w \in \pi_1(A_\alpha \cap A_\beta)$ then $(\iota_{\alpha\beta})_\star(w) \left((\iota_{\beta\alpha})_\star(w)\right)^{-1}$ is in the kernel of Φ.

Let N denote the normal subgroup of $\star_\alpha \pi_1(A_\alpha)$ generated by such elements. Then $\ker(\Phi) = N$ (under assumptions).

Theorem 8.8 (Seifert-Van Kampen Theorem)**.** *If X is a union of path-connected open sets A_α such that $x_0 \in X$, and $x_0 \in A_\alpha$ for all α, and $A_\alpha \cap A_\beta$ is path-connected, and $A_\alpha \cap A_\beta \cap A_\gamma$ is also path-connected for all α, β, γ. Then Φ induces an isomorphism*

$$\pi_1(X) \cong \star_\alpha \pi_1(A_\alpha)/N$$

Proof. We have shown before that Φ is surjective. Now we will show that $\ker(\Phi) = N$. Define a factorization of an element $[f] \in \pi_1(X)$ as the product $[f_1] \dots [f_k]$ such that each $[f_i] \in \pi_1(A_\alpha)$ for some A_α and $f \sim_p f_1 \star f_2 \star \cdots \star f_k$. A factorization of f is thus a word in $\star_\alpha \pi_1(A_\alpha)$. We will say that any two such factorizations are equivalent if they are related by the following two moves or their inverses:

(1) Combine adjacent terms $[f_i][f_{i+1}]$ into a single term if $[f_i][f_{i+1}]$ are in the same group.
(2) If $f_i \in A_\alpha \cap A_\beta$ we may regard $[f_i] \in \pi_1(A_\alpha)$ as $[f_i] \in \pi_1(A_\beta)$.

The first move does not change the factorization. The second move does not the element in $\star_\alpha \pi_1(A_\alpha)/N$.

Let us assume that $[f_1] \dots [f_k]$ and $[f_1'] \dots [f_l']$ are two factorizations of $[f] \in \pi_1(X)$. We will show that they are equivalent. We have,

$$f_1 \star \cdots \star f_k \cong f_1' \star \cdots \star f_l'$$

That is, there exists a path-homotopy $F : I \times I \to X$. Now choose a partition of $I \times I$ by

$$0 = s_0 < \cdots < s_m = 1 \text{ and } 0 = t_0 < \cdots < t_n = 1$$

such that each rectangle $R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped to a single A_α by F. We label this A_α by A_{ij}.

Now we will modify the partition in two ways:

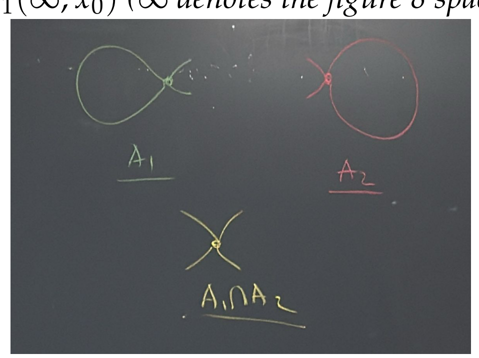
(1) Add subdivisions to the top ($[t_{n-1}, t_n]$) and bottom row ($[t_0 - t_1]$) such that for each end point in the partition there exists a corresponding s_i.
(2) Have at least 3 rows.
(3) Every point in $I \times I$ must lie in at most 3 (smaller) rectangles
(4) Rename the smaller rectangles (left to right, bottom to top) using R_i.

Define γ_r as the path from the left to the right edge of $I \times I$, and it separates rectangles $R_1, \dots R_r$ from the rest of the rectangles.

Note that F_{γ_r} is a loop based at x_0. And all F_{γ_r} are path homotopic. Then $F|_{\gamma_r}$ has a factorization. If $F(v) \neq x_0$, then we have a path from $x_0 \to F(v) = F|_{\gamma_1}$ □

Example 8.2.

(1) $\pi_1(\infty, x_0)$ (∞ denotes the figure 8 space).



Then, $\pi_1(\infty, x_0) \cong (\pi(A_1) \times \pi_1(A_2))/N = (\mathbb{Z} \times \mathbb{Z})/N$. *Since* $N = A_1 \cap A_2$ *deformation retracts to a point this is* $\mathbb{Z} \times \mathbb{Z}$.

(2) $\pi_1(S^1 \wedge S^2) \cong \mathbb{Z}$

$X_1 \wedge X_2 := (X_1 \sqcup X_2)/(x_1 \sim x_2)$ *for some* $x_1 \in X_1$, $x_2 \in X_2$.

(3) $\pi_1(\mathbb{R}^3 \setminus S^1)$ ***TODO:***

9. Cell Complexes

Definition 9.1. *We start with a discrete space whose points are regarded as* 0*-cells. Now, we inductively define the* n***-skeleton*** X^n *from* X^{n-1} *by attaching* n*-cells* e^n_α *via maps* $\phi_\alpha : S^{n-1} \to X^{n-1}$ *where* e^n_α*s are* D^n_α. *Hence,* $X^n = X^{n-1} \sqcup D^n_\alpha/(x \sim \phi_\alpha(x)$ *if* $\alpha, \forall x \in S^{n-1})$. *We say a topological space* X *has a CW/CX structure if* $X = \cup_n X^n$.

Definition 9.2. *Each cell* e^n_α *of a CW CX has a map* $\Phi_\alpha : D^n_\alpha \to X$ *which extends* $\phi_\alpha : \partial D^n_\alpha \to X$. *It is defined by*

$$D^n_\alpha \hookrightarrow X^{n-1} \sqcup D^n_\alpha \to \underset{\phi_\alpha : \partial D^n_\alpha \to X^{n-1}}{X^{n-1} \sqcup D^n_\alpha} = X^n \hookrightarrow x$$

This map is called the ***characteristic map*** *asssociated to* ϕ_α.

Definition 9.3. *A* ***subcomplex*** A *of a CW CX* X *is a subspace* $A \subset X$ *such that* A *is a union of some number of cells in* X *with a attaching map for each of those cells has a image contained in* A. *We say that* (X, A) *is a CW pair.*

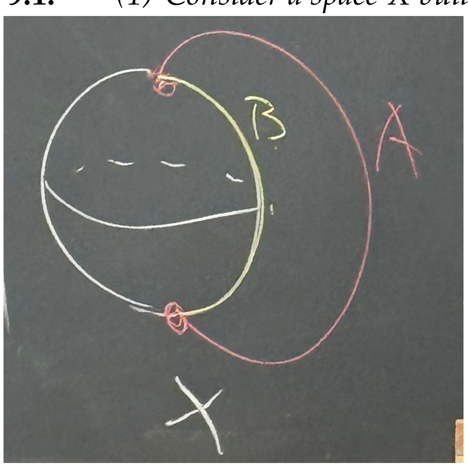
Definition 9.4 (Quotinent CX)**.** *If* (X, A) *is a CW pair, then* X/A *inherits a canonical cell structure from* X. *The cells of* X/A *are the cells* $X - A$, *plus one* 0*-cell given by the image of* A *in* X/A. *For a cell* e^n_α *of* $X - A$ *with* $\phi_\alpha : S^{n-1} \to X^{n-1}$ *, the corresponding attaching map in* X/A *is*

$$S^{n-1} \to X^{n-1} \to X^{n-1}/A^{n-1}$$

Theorem 9.5. *If* (X, A) *is a CW pair and* A *is contractible then* $q : X \to X/A$ *is a homotopy equivalence.*

Proof. Hatcher Chapter 0, Homotopy Extension Lemma $\square$

Example 9.1. (1) *Consider a space* X *built from* S^2 *by attaching an arc to two points in* S^2.



Both (X, A) *and* (X, B) *are CW complexes.*

(2) $\pi_1(S^3, S^1) =?$ **TODO:**

10. Homology

Remark 10.1. *$\pi_n(S^n) = \mathbb{Z}$, and $\pi_m(S^n) = 0$ if $m >\leq n$. We don't know how to compute $\pi_m(S^n)$ for $m > n$.*

Homology distinguishes higher dimension space and it is easy to compute. We define simplical and singular homology separately, but eventually they have the same properties.

10.1. Simplical Homology.

Definition 10.2. *A **n-simplex** is the smallest convex set in $\mathbb{R}^m$ spanned by $(n+1)$ points $v_0, \ldots, v_n$ such that all of them do not lie in any hyperplane of dimension less than n. That is, $v_1 - v_0$, $v_2 - v_0, \ldots,$ $v_n - v_0$ are linearly independent. We say $(v_0, \ldots, v_n)$ comes with the order $v_0, \ldots, v_n$.*

The standard n-simplex is

$$\left\{ (t_0, \ldots, t_n) \in \mathbb{R}^{n+1} \mid \sum_i t_i = 1, t_i \geq 0 \right\}$$

There is a canonical homeomorphism from Δ^n to any n-simplex.

$$(t_0, \ldots, t_n) \mapsto \sum t_i v_i$$

Definition 10.3. *Δ-complex structure of a space X is a collection of maps $\sigma_\alpha : \Delta^n \to X$.*

1. *$\sigma_\alpha|_{\Delta_n^\circ}$ is injective and each point of X is in the image of exactly one such $\sigma_\alpha|_{\Delta_n^\circ}$.*
2. *$\sigma_\alpha|_{\text{face of } \Delta_n}$ is one of the maps $\sigma_\beta : \Delta^{n-1} \to X$, where a face of Δ^n is a $(n-1)$-simplex that we get by deleting a vertix of Δ^n.*
3. *A set $U \subset X$ is open iff $\sigma_\alpha^{-1}(U)$ is open in Δ^n, for all α.*

Definition 10.4. *Let X be a Δ-complex and let $\Delta_n(X)$ denote the free abelian group with basis n-simplicies of X. The elements of $\Delta_n(X)$ are of the form $\sum_i n_i \sigma_i$ where this is a finite sum, $n_i \in \mathbb{Z}$ and $\sigma_i : \Delta^n \to X$.*

We denote the deletion of v_i from the simplex as $[v_0, \ldots, \widehat{v}_i, \ldots, v_n] \cong [v_0, \ldots, v_{n-1}]$. We will always orient the 1-simplex $[v_i, v_j]$ by an arrow going from v_i to $v - j$ for $i < j$.

Proposition 10.5. *The composition*

$$\Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \Delta_{n-2}(X)$$

is zero ($\partial_{n-1} \circ \partial_n = 0$).

Proof.

$$\partial_n(\sigma) = \sum_i (-1)^i \sigma|_{[v_0,\ldots,\widehat{v}_i,\ldots,v_n]}$$

$$\partial_{n-1} \circ \partial_n(\sigma) = \sum_{j<i} (-1)^i(-1)^j \sigma|_{[v_0,\ldots,\widehat{v}_j,\ldots\widehat{v}_i,\ldots,v_n]} + \sum_{i<j} (-1)^i(-1)^{j-1} \sigma|_{[v_0,\ldots,\widehat{v}_i,\ldots\widehat{v}_j,\ldots,v_n]}$$

□

Definition 10.6. *In general, whenever we have* (C_n, ∂_n)

$$\to C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}}$$

with $\partial_n \circ \partial_{n+1} = 0$, *then* $\operatorname{img}(\partial_{n+1}) \subset \ker(\partial_n)$. *Hence we define the n-th homology group,*

$$H_n := \ker(\partial_n) / \operatorname{im}(\partial_{n+1})$$

Hence the n-th simplicial homology of X is $H_n^{\Delta}(X) = \ker(\partial_n) / \operatorname{im}(\partial_{n+1})$.